

Chapter 7

De-adjectival and de-“adjectival”: Greek κρύπτω ‘cover’ and Proto-Germanic *χreuðaną ‘deck, equip, adorn’

Andrew Merritt

Georgetown University

This article clarifies the relationships of three secondary roots reflected in, e.g., 1) Greek κρύπτω ‘cover’ and Tocharian B *kraup-* ‘gather, amass’ (: **kreyb^h-*), 2) Old Church Slavonic *kryti* ‘cover’ and Lithuanian *kráuti*, 1sg. prs. *kráuju* ‘pile up’ (: **kreyh_x-*), and 3) Old English *hréodan** ‘deck, adorn’ (: **kreyd^h-*), in light of our current understanding of the derivational behavior of nominal compounds formed on the basis of the predicative instrumental periphrasis. After identifying **kreh₂-* ‘heap, strew, cover’ (Latvian *krāju*, *krāt* ‘put aside, accumulate, amass’) as the primary root, the author proposes that its *u*-stem derivative **kró/éh₂-u-* ‘heap, covering’ was the base of an *-*ie/o*-present **kreh₂-u-*ie/o-** ‘provide with covering’, whose reanalysis as a primary verb gave rise to **kreyh₂-*. Since the original denominal stem could only appear in the present system, there was need elsewhere for the use of its base’s instrumental with the light verbs **d^heh₁-* ‘put, make’ and **b^huh_x-* ‘become’. On the basis of the anticausative periphrasis **kréh₂-u-h₁ b^huh_x-* ‘become covered’ (cf. Ved. *gúhā bhū-* ‘become hidden’) was formed an adjective **kṛh₂-u-b^h(u)-ó-* ‘heaped, covered’ >> **kruh₂(-)*b^h-ó-** basic to a **kruh₂b^h-*ie/o-** ‘render covered, heap (upon)’, whose reflex **krub^h-*ie/o-** was the preform of κρύπτω and the model for the **kreyb^h-* ‘heap’ ancestral to the Tocharian root. On the basis of the causative periphrasis **kréh₂-u-h₁ d^heh₁-* was formed a **kṛh₂-u-d^hh₁-ó-* >> **kruh₂(-)*d^hh₁-ó-**, whose resultative patientive interpretation ‘covered’ paved the way for the quasi-Property Conceptual meaning ‘equipped, adorned’. When this adjective formed an *s*-stem substantive **kréud₂(-)*d^hh₁-o/es-** ‘equipment, adornment’ > **kréud^hh₁-o/es-*, this quasi-PC meaning and the frequency of *s*-stems beside simple thematic presents, both prototypically radical *é*-grade categories, contributed to the formation of a causative **kréud^hh₁-e/o-* ‘render equipped, adorned’ derived by suffixal substitution from a new effectively adjectival root.



1 Introduction: from specific to general

The purpose of this contribution is, on the one hand, to apply a set of theories to explain precisely how the members of a specific set of Indo-European data are historically related and, on the other, to highlight how such explanations offer theoretical implications for our general understanding of various processes whereby verbal stems are formed on the basis of adnominals,¹ whether nominal stems or semantically adjectival² roots. It will be argued that this crucial distinction between the two types of de-adjectival verbs, namely, stem-derived versus root-derived, is reflected in the prehistory of, most notably, Greek κρύπτω ‘cover, hide, bury’ and Old English *hréodan** ‘deck, adorn’, which have been assumed to be related in terms of root-etymology (IEW 616–617), but whose exact derivational prehistory is unclear. The goal of this paper is to clarify that history by (1) identifying the primitive root as **kreh₂-* (LIV² 367), which seems to have meant ‘cover, heap’; (2) proposing that a *u*-stem nominal derivative of that root was used in the predicative instrumental periphrasis with the light verbs **d^heh₁-* ‘put, make’ (causative) and **b^huh_x-* ‘become’ (anticausative); and (3) arguing for the derivation, on basis of the respective periphrases, of adjectival nominal compounds whose own derivatives illustrate the interplay of morphology and lexical semantics in de-adjectival derivation.

Our set of data, derived largely from the material under Pokorny’s **krā[u]-* : *krəu-* : *krǔ-* ‘aufeinander, auf einen Haufen legen, zudecken, verbergen’ (IEW 616–617), includes Greek κρύπτω ‘cover, hide, bury’ (: κρύφα ‘secretly’ : κρυφ-), Tocharian B *krauptār* ‘gathers’ (TA *krop-* ‘id.’), OCS *kryjo*, *kryti* ‘cover’, Lithuanian *kráuju*, *kráuti* ‘pile up’ (Latv. *kraūju*, *kraūt* ‘id.’), and Old English *hréodan** ‘deck, adorn’ (pret. *hréad* : pret. part. *gehroden*, ON *hrjóða** ‘id.’ : *hroðinn* ‘painted’, *gull-roðinn* ‘gilt’). Given that all the above-listed data bear segments reflecting the sequence **kru-* and convey lexical meanings involving the notion of covering or placement together, it remains appropriate to maintain that Pokorny is basically correct in supposing that these Greek, Tocharian, Slavic, Baltic, and Germanic forms ultimately share a root reflected in a series of roots reconstructible as secondary. The question that confronts us if we accept this supposition is the origin of the material following the reflexes of **kru-* in each subset.

¹I use “adnominal” in a general sense to refer (with the exception of finite verbs) to any element attributed to or predicated of a substantive.

²For the derivational behavior of such roots, see Nussbaum 2022.

2 The existence of **kreub^h-*, **kreuh_x-*, and **kreud^h-*

The first consists of Greek κρύπτω and Tocharian B *kraup-* (: *kraupe* m. ‘group’), the latter of which, since it forms a class II present, a class II subjunctive, and a class I preterite (Adams 2013: 236–238), is likely, in light of the oddity of that pattern, to be inherited (cf. athematic Tocharian A *kroptār* class I pres., 3pl. *kropāntār*, Malzahn 2010: 614–616). What this comparison allows us to infer is, therefore, that κρύπτω and *krauptār* reflect derivatives of a secondary root at least as old as Proto-Nuclear Indo-European, namely a **kreub^h-* ‘heap, strew, cover’ which formed, in descriptive terms, a radical zero-grade **-ie/o-* present **krub^h-ie/o-* ancestral to κρύπτω as well as what appears to be a radical *ó*-grade uncharacterized present ancestral to *krauptār* (< Transponat **króub^h-e-tor*). Even if we attribute this root to PIE, it is not, in accordance with our hypothesis, a primary product of reconstruction, as one would assume for a lemma in LIV². To deal with our second subset, which includes OCS *kryti* ‘cover’ and Lithuanian *kráuti* ‘pile up’, LIV² reconstructs a **kreuh_x-* ‘aufhäufen, bedecken’ (LIV² 371), from which were formed, in descriptive terms, the **kruh_x-ie/o-* ancestral to *kryjō* and the **kreuh_x-ie/o-* indirectly ancestral to *kráuju*³ (cf. dial. Lith. *kriáuju* : Latv. *kraūju*).

While LIV² ignores κρυφ- and *kraup-* encountered in Pokorny, it does take account of the Germanic data via the assumption that OE *hréodan** ‘bedecken’ reflects a **-d^he/o-* present **kréuh_x-d^he/o-*. Though we have good reason to believe that there was indeed a morphological boundary before that dental segment at some historical stage, it appears premature to characterize *hréodan** as the reflex of a present in **-d^he/o-*. Unlike, for example, βαρύθω ‘get weighed down (intr.)’, which is formed, with the reflex of a descriptive **-d^he/o-*, from the actual stem of an adjective (βαρύς ‘heavy’), the Proto-Germanic **χreuðanq* ‘deck, equip, adorn’ ancestral to *hréodan** behaves as a root-based formation parallel to other derivatives of its root (: **χrauðō-* [ON *hrauð* f. ‘chainmail shirt’, OE *earm-hréad* ‘arm-ornament’] : **χrusti-* [OHG *hrust* ‘armor’, OE *hyrst* ‘decoration, armor’]). In this formal respect, **χreuðanq* rather resembles βρῖθω ‘become weighty’, which, though historically related to βαρ- ‘heavy’ (v. Rasmussen 1989: 94–95), is formed from its own root (: βρῖθ-ύς ‘heavy’, βρῖθ-ος ‘weight’). While it is possible to claim that the **-ði/a-* of **χreuðanq* reflects an erstwhile presential suffix that formed a stem eventually reanalyzed as a (secondary) **kreud^h-* ‘cover’, such a claim is of no profit in the absence of an attempt to explain the **kreud^h-* in question, which must have existed at some stage, in light of any explanations of the origins of the **kreub^h-* and **kreuh_x-* to which it is assuredly related.

³On the acute intonation, see Villanueva Svensson 2014.

3 Evidence for primary **kreh₂*- ‘heap, strew, cover’

The initial clue serving in an explanation of how the three secondary roots, **kreub^h*-, **kreuh_x*-, and **kreud^h*-, are related is in fact contained in Pokorny’s entry. The brackets of his **krā[u]*-, the first-listed variant of the root, betray the inkling that the truly primary form of the root did not include the **-u*-. Especially in light of the Latvian *krāju*, *krāt* ‘put aside, accumulate, amass’ listed under **krā[u]*-, it is deeply appealing to reconstruct the truly primary form as a **kreh₂*- ‘heap, strew, cover’, which LIV² 367 quizzically represents as a *?*kreh₂*- ‘aufhäufen, sammeln’ supposedly reflected solely in Balto-Slavic. According to Pokorny, one may group *krāju*, *krāt* together with OCS *kradŏ*, *krasti* ‘steal’ “mit präsensbildendem d” and, with an apparent *p*-extension, Lithuanian *krópti* ‘id.’ and Latvian *krāpju*, *krāpu*, *krāpt* ‘cheat, deceive’. Though the origin of such formants or extensions remains to be determined, what is clear is the likelihood that these verbs, given their senses ‘steal’ or ‘deceive’, should be related to the one meaning ‘amass’. Since accumulation generally involves the placement or position of a substance or set of objects over and upon an entity, and since the entity positioned over and upon another entity may serve to effect the latter’s non-stimulus status (e.g., soil of a mound covering a dead body and/or treasure), it is likely that the senses of theft and deception proper to *krasti*, *krópti*, and *krāpt* developed via ‘hide, conceal’ from ‘heap, strew, cover’ (cf. PDE *bury*; OIr. *arcelim* ‘steal’ : *ceilid* ‘hides’, Buck 1949: 791).

Some evidence for this original sense of accumulation proper to **kreh₂*- may also be adduced from Greek. Beside evidence of a κλώμαξ, -ακος, m. ‘heap of stones, rocky place’ and a κλωμακόεις ‘rocky’ (B 729), Hesychius attests to a κρώμαξ· σωρὸς λίθων and a κρωμακόεν· κρημνῶδες. While Beekes (EDG 720), rejecting the association of the former set with κλάω ‘break’, considers the variation between -λ- and -ρ- to be evidence that both sets are of non-Indo-European origin, an evaluable hypothesis would take them back to two PIE roots. Before considering the question of the roots themselves, we should first observe that the sequence -μακ- points to derivation from stems in -μα (cf. ἔρμαξ, -ακος f. ‘heap of stones, cairn’ ← ἔρμα ‘prop, support’ [Hom.]). By comparison to Latin *rūpēs* f. ‘rock’ (: *rumpō* ‘break’), it is probable that, in the case of κλώμαξ, the μα-stem in question was indeed akin to κλάω. If, however, the earlier meaning of that μα-stem was ‘heap’, it is not inconceivable that its preform was derived from **kleh₂*- ‘heap up, put’ (**k^(u)leh₂*- ‘hinbreiten, hinlegen’, LIV² 362), evidence of whose verbal derivatives, namely Lithuanian *klóju*, *klóti* ‘spread, lay down’ (Latv. *klāju*, *klāt* ‘id.’) and OCS *kladŏ*, *klasti* ‘load, lay’, incidentally resembles that of **kreh₂*- ‘heap, strew’. In the case of the base of κρώμαξ, it is fairly certain that its meaning was ‘heap’.

Two additional data deserve mention: Κρῶμνα, -ης (: πόλις Παφλαγονίας B 855 [Hsch.]; a toponym) and the reportedly Paphlagonian κρωμακωτός ‘rocky and steep’. Given the identity of place as well as toponomastic practice (cf. Αἵπεια ‘Hightown’ : αἰπός, αἰπός ‘high, lofty’), it is likely that the base of the *deví-* derivative ancestral to Κρῶμνα was reflected as the μα-stem basic to the κρῶμαξ from which κρωμακωτός was derived. In all likelihood, this μα-stem would go back to a *-men- stem derived from **keh*₂- ‘heap, strew’.

Comparable in form to the **póh*₂-*mē* ‘protection’ (: **peh*₂- ‘cover, protect, care for’) ancestral to πῶμα ‘lid, cover’ (Hom.),⁴ this *-men- stem would have been an **ó/é* acrostatic. In addition to the association between κρῶμαξ and **keh*₂- effected by the attribution of the *-men- stem to this type, the reconstruction of a **króh*₂-*mē*/*kréh*₂-*mē*- ‘heaping, heap’ has the potential advantage of providing a way to account for the origin of κρημνός ‘overhanging bank, cliff’. Since the appearance of its -η- in Pindar appears to preclude an *-ā- in the preform (EDG 777), it is conceivable that κρημνός goes back to a genitival **kréh*₂-*mn-ó*- ‘of the heaping, heaped’ derived from **króh*₂-*mē*/*kréh*₂-*mē*- ‘heaping, heap’. Though this scenario has some problems,⁵ it does provide κρημνός with a semantically straightforward etymology (cf. Hitt. *wappu*- ‘riverbank’ < *(*h*₂)*uóp-u-/h*₂*uép-u*- ‘heap’ : **h*₂*uēp-* ‘cast, strew’ [Hitt. *huwapp-/hupp-* ‘cast, throw’ : Ved. *vápati* ‘strews’], Catsanicos 1985: 125; Melchert 2013: 176) whose formal features are mostly in line with a simple thematic item conceivably derived from a *-men- stem to **keh*₂- ‘heap, strew’.

By comparison to the form of Latin *fās* ‘morally right’ (< **b^héh*₂-*o/es-* ‘saying, utterance’, EDL 203), we may hypothesize that **keh*₂- also formed an s-stem substantive **kréh*₂-*s-* ‘heaping, strewing, spreading, covering’ for which there is some indirect evidence in Baltic. Beside Lithuanian *krúsnis*, -*ies* f. ‘pile of stones in a field’ (v. Smoczyński 2018: 616), identifiable as the reflex of a **kruh*₂-*s-n-i*- derived from a possessive in *-*nó-* to an s-stem **kréh*₂-*o/es-* ‘heap, covering’ to be discussed below (: **króu*_h*h*₂-*o-* ‘covering’ [> OCS *krovъ* ‘roof’]), Lithuanian and Latvian respectively attest to a *krósnis*, -*ies* f. ‘(stone-)stove; pile of stones’ and *krāsns* ‘stove’, which Smoczyński (2018: 613) and Karulis (1992: 419) both connect with Latvian *krāju*, *krât* ‘amass’. While Smoczyński’s connection with *krât* and thus **keh*₂- is entirely justified, what this scholar analyzes as a suffix -*sni-* is in need of further clarification. If we compare the dialectal Latvian *krāsts* m. ‘oven’ (: *krāstīt* ‘collect, hoard’), we may suppose a formation similar to Latin *fāstus*

⁴Analysis of πῶμα ‘lid, cover’ as the reflex of an **ó/é* acrostatic derives from Nussbaum (p.c.).

⁵One would have to assume that this derivation took place at a very early date in the protolan-guage and that one of the nasals failed to be deleted as expected or was restored.

‘right’ (← *fās* ‘id.’), that is, a possessive in **-tó-* derived from an *s*-stem basic to a **-nó-* stem from which the ancestor of this *-sni-* stem was formed. Since that *s*-stem had the sense ‘heaping, heap’, its derivative, namely **kṛh₂-s-nó-* (cf. λύχνος ‘lamp’ < **lúk-s-no-* ‘the lighted’ ← **luk-s-nó-* ‘lighted’ ← **léuk-o/es-* ‘light’), would have borne a meaning ‘heaped’ subsequently determined by means of the acrostatic *i*-stem derivative **kréh₂-s-n-i-* ‘the heaped, heap’ ancestral to the Baltic cognates. Since the above nominal and verbal evidence supports the reconstruction of **kreh₂-* as a primary root, the immediate question is the nature of the **-u-* in the secondary roots. As the old extensions invoked by Pokorny have increasingly ceded to the discipline’s growing powers of morphosemantic analysis, we can assume that the **-u-* is of suffixal origin.

4 The nature and derivational role of **kró/éh₂-u-* ‘heap, covering’

4.1 Background

This assumption opens up two paths of hypothesis: (1) it is a verbal stem formant, in which case it would be the suffix of a *u*-present, or (2) it is the suffix of a nominal stem. The preferability of the latter hypothesis becomes clear in light of parallelism with the derivational history assigned in previous research to κλύπτω ‘cover’ (Merritt 2021). This particular history derives its existence from the understanding, originating in and enriched by, respectively, comparative-historical (Jasanoff 1978: 118–126; 2004; Schindler 1980; Balles 2006; Rau 2009) and theoretical-typological work (Dixon 1982, 2004; Francez & Koontz-Garboden 2015), that PIE was a language in which certain stative concepts, such as Property Conceptual (PC) states, were disposed to realization via substantives in the instrumental case. In prelinguistic terms, a stative concept could be accessed by conceiving of a substance, whether concrete or abstract, as an entity possessed by the being of which the state is predicated. While much attention has been paid to the role of adjectival (i.e., PC) abstracts in this means of stative predication, we should not forget that the specific stative/anticausative/causative⁶ construction which inaugurated much of this research tradition, namely Vedic *gúhā as-/bhū-/dhā-* ‘be/become/make hidden’, includes the reflex of a substantive derived ultimately from a root that basically meant ‘cover’ (Lith. *gũžti* ‘[of hens] sit,

⁶I use such terms in a general sense (cf. introduction to this volume: 14–16). Accordingly, a causative and anticausative derived from an adjective or adjectival root are specifically factitive and fientive respectively.

squat, keep warm’), corresponding to an Activity Concept, rather than a Property Concept. Though *gúhā* in the sense ‘hidden, secret’ does express (approximately) a PC state, its preform, in light of the metonymy COVERED FOR HIDDEN underlying that sense, must have resembled a verbal resultative state, namely ‘covered’ (lit. ‘with covering’), predicated of a patient/theme. In like manner, there is, at the heart of the history of *καλύπτω*, an acrostatic *u*-stem substantive **kó/él-u-* ‘covering’, whose instrumental was used to predicate just such an Activity Concept state. Correspondingly, it seems feasible to suppose that our **kreh₂-* was ultimately the base of a **kró/éh₂-u-* ‘heap, covering’.

4.2 **kreh₂-u-je/o-* ‘provide with a covering’

In accordance with this hypothesis, it is conceivable that the instrumental of **kró/éh₂-u-*, namely **kréh₂-u-h₁*, was used as an adnominal in nominal sentences or (anti)causative periphrases with forms of **b^huh_x-* ‘become’ and **d^heh₁-* ‘put, make’. While the instrumental form would participate in periphrasis, it is also conceivable that the meaning of the instrumental could be expressed without the morphology of that case by means, for instance, of the suffix **-je/o-*. As a denominal verbal stem formant, **-je/o-* lets the constructional and lexical semantic idiosyncrasies of the base shape the meaning of what it derives. In principle, it does not strictly specify the quantity or quality of the arguments. In practice, however, there seems to be a tendency for transitivity and for the agentivity of the argument serving as subject. For example, the derivational semantic history of Latin *arguō* ‘demonstrate’ is discernible by supposing that it reflects a transitive **-je/o-*-present **h₂erġ-u-je/o-*⁷ ‘[x] provide [[y] [with clarity]]’ derived from **h₂ó/érġ-u-* ‘clarity’ in the sense corresponding to its instrumental **h₂érġ-u-h₁* ‘with clarity’, from which in theory was formed a *t*-stem (abstract)⁸ substantive basic to the possessive adjective ancestral to the “presuffixally lengthened” perfect passive participle *argūtus* ‘demonstrated’ (< Transponat **h₂erġ-u-h₁-t-o-*; Nussbaum 1996; cf. EDL 53). Given the frugality of this analysis, it is appealing to suppose that **kró/éh₂-u-* ‘heap, a covering’ was the base of an instrumentative PIE **-je/o-*-present **kreh₂-u-je/o-* ‘provide with a covering’.

⁷The cognation of Hittite *arkuwaē^{-zi}* ‘make a plea’, which Melchert 1998 associated with *arguō* (< **argu-yé/ó-*), would be difficult to maintain, unless one assumes on the basis of Kloekhorst 2006 and 2008: 206 that, since the root **h₂erġ-* of the form ancestral to *arkuwaē^{-zi}* was in *o*-grade (rather than the *e*-grade found elsewhere), the initial *h₂* was lost.

⁸Cf. Hitt. *happina-* ‘rich’ → abstract *happinatt-* (c.) ‘wealth’, Nussbaum 2004.

4.3 The genesis of **kreuh₂-* ‘heap, cover’

Since this verb’s meaning was equivalent to that of the verb originally derived from **kreh₂-* ‘heap, strew, cover’, it is conceivable that it was reanalyzed as a primary present derived from a “long-diphthongal” **kreh₂u-* ‘heap, strew, cover’. Accordingly, the root of this present would have appeared eligible as a base to a **-tó-* verbal adjective, namely a **kṛh₂u-tó-* ‘heaped, covered’. Though not of the canonical structure reflected in Vedic *pītá-* ‘drunk’ (< **pih₃-tó-* < **ph₃(-i)-tó-*), this **kṛh₂u-tó-* may be assumed to have undergone laryngeal metathesis in the same way as the preform of Latin (← Sabellic) *brūtus* ‘heavy, stupid’, which via **g^hruh₂tó-* would reflect, whatever its exact analysis, a **g^hṛh₂-u-t(-)ó-* ‘heavy’ related to the preform of βαρύς ‘id.’. The resulting **kruh₂-tó-* ‘heaped, covered’, arguably basic to the preform of Lithuanian *krūtis* f. ‘pile of stones, stone-oven’ (< **kruh₂-t-i-*) and directly ancestral to the past passive participle of OCS *kryti* ‘cover’ (cf. Russian *krýtyj* ‘covered’), appears to have been pivotal in the creation of what we may now rewrite as our secondary **kreuh₂-* ‘heap, cover’ (→ τόμος-substantive **króuh₂-o-* ‘covering’ [> OCS *krovŭ* ‘roof’] : s-stem **kréuh₂-o/es-* ‘heap, covering’ → **kruh₂-s-ó-* ‘heaped, cover(ed)’ [→ **króu(h₂)-s-o-* → **króu(h₂)-s-ijo-* > ON *hreyši* ‘stone heap’] → **kruh₂-s-ijo-* → **krúh₂-s-iĵe-h₂-* > Russian *krýša* ‘roof’). While Slavic points to a radical zero-grade **kruh₂-ĵe/o-* derived from **kreuh₂-* on the model of **kruh₂-tó-*, Baltic inherited a radical *e*-grade **-ĵe/o*-present **kreuh₂-ĵe/o-* ‘heap’ (>> dial. Lith. *kriáuju* : Latv. *kṛāŭju*) that arose conceivably by analogy to families involving (as usual) radical zero-grade **-tó-* verbal adjectives beside radical *e*-grade **-ĵe/o*-presents (e.g., **ten-ĵe/o-* ‘stretch’ [τείνω, Alb. *n-den* ‘stretches out’, LIV² 627] : **tṇ-tó-* ‘stretched’ [Ved. *tatá-* ‘id.’, τατός ‘extensible’]).

5 The origin of κρύπτω

5.1 Background

To understand the emergence of **kreuh^h-* beside the **kreuh₂-* whose origin we have just considered, we should return to the original instrumentative **kreh₂-u-ĵe/o-* ‘provide with a covering’. Since, by its very nature, this denominal was a creature of the present system, the non-present system had to be served by periphrases consisting of the instrumental of the denominal’s base and the light verbs **d^heh₁-* and **b^huh_x-*. Given the necessity of periphrasis at some level, it is conceivable that the option to form compounds based on that periphrasis was realized. We may thus suppose that the anticausative phrase **kréh₂-u-h₁ b^huh_x-*

‘become covered’ inspired a resultative compound adjective **kṛh₂-u-b^huh_x-ó-* ‘(having) become heaped, covered’, whose νεογνός-reflex was a **kṛh₂-u-b^h(u)-ó-*. This form was ancestral to an eventually simple thematic adjective **kruh₂b^h-ó⁹* ‘heaped, covered’ via laryngeal metathesis and morphological reanalysis. As a thematic adjective, this **kruh₂b^h-ó-* would in theory have been eligible as a base to a *nēwahhi*-type factitive. What we find instead is the reflex of a **-je/o*-present which, like χαλέπτω ‘treat harshly’ beside χαλεπός ‘harsh’, was derived by substitution of **-je/o-* for the thematic vowel of the adnominal base. The derivative would thus have been **kruh₂b^h-je/o-* ‘render covered, heap (upon)’.

5.2 The parallel of πλάσσω ‘form, mould’

The morphological and phonological nature and fate of this denominal may be compared to the history of πλάσσω ‘form, mould’, which, given the sequence πλαθ- (: κοροπλάθοις ‘figure-modeler’), Pokorny (IEW 805–807) characterizes as a **-je/o*-present formed from a base containing the remnant of a *d^h*-formant of a present stem derived ultimately from what would now be represented as **pleh₂-* ‘broad, flat’ (Hitt. *palhi-/palhai-* ‘wide, broad’, Lat. *plānus* ‘flat’, OE *flōr* f. ‘floor’). Since, however, πλάσσω does reflect a present in **-je/o-* (PGk. **plat^h-je/o-*), it is preferable to analyze the preform of this verb as denominal, specifically de-adnominal from a **-d^hh₁-o-* compound. In maintenance of the connection with the descriptive **pleh₂-* ‘broad, flat’, such an analysis would begin with the derivation of what may be provisionally represented as a root noun **pléh₂-/plh₂-* ‘the flat, flatness’, whose capacity to modify substantives rendered it eligible for use in the familiar predicative instrumental periphrasis (i.e., the *cale-faciō*-type). Accordingly, on the basis of a causative phrase **plh₂-éh₁ d^heh₁-* ‘provide with the flat, flatten’ was formed **plh₂-d^hh₁-ó-* ‘flattened, pressed, kneaded, moulded’, apparently basic to the substantive **plh₂(-)-d^hh₁-e-h₂-* ‘the moulded, mould’ ancestral, via the παλάμη- or *palma*-rule (Rix 1996: 157–158, Meiser 1998: 108–109, Höfler 2017, Weiss 2020: 119, Hackstein 2021: 193–194), to the παλάθη ‘cake of preserved fruit’ adduced by Pokorny (‘flacher Fruchtkuchen’; cf. παλάθαι· σύκων μαζία [Hsch.] : μάσσω ‘knead, mould’).

The sequence πλαθ- brings up the question of the origin of the radical base πλάθ- underlying πλάσσω and apparent in its nominal relatives (e.g., πλάθανον ‘baking-mould’, πλαθά f. ‘modelled figure’). Given that πλάθ- does not display a laryngeal reflex, it is necessary to suppose that speakers posited the root on the basis of a form in which the laryngeal was deleted. Since the substitution

⁹For the PIE status of the development **-b^hu-* to **-b^h-*, see Merritt 2021: 312–313.

of $*\text{-}\dot{\imath}e/o\text{-}$ for the thematic vowel of our $*p_lh_2(-)d^h h_1\text{-}\acute{o}\text{-}$ would have produced a sequence subject to Pinault's Law and, subsequently, the *Weather Rule* (cf. Ion. γλᾶσση- 'tongue' < $*g_lh_3g^h\text{-}\dot{\imath}eh_2\text{-}$ 'pointed' [cf. γλωχίς 'point, barb'] : $*gl(e)h_3g^h\text{-}\dot{\imath}h_2\text{-}$ [: γλωττα 'tongue']; Neri 2017; Weiss 2020: 123), it is appropriate to suppose that $*p_lh_2(-)d^h h_1\text{-}\dot{\imath}e/o\text{-}$ 'get x moulded' was reflected as an eventual $*p_l d^h\text{-}\dot{\imath}e/o\text{-}$ directly ancestral to πλάσσω, whose meaning 'form' resulted from a metonymic extension on the part of $*p_lh_2(-)d^h h_1\text{-}\acute{o}\text{-}$ from 'flattened, pressed' to 'kneaded, moulded' via the notion of applying pressure to change or create a surface. Morphologically, the result is clear: the $*\text{-}\dot{\imath}e/o\text{-}$ -present $*p_l d^h\text{-}\dot{\imath}e/o\text{-}$ was reanalyzed as primary from a $*pled^h\text{-}$ 'form, shape' that could now form derivatives of its own, such as the τόμος-type $*plód^h\text{-}o\text{-}$ 'formation' arguably ancestral to OCS *plodъ* 'fruit, fetus' (cf. $*kreu\dot{u}h_2\text{-}$ 'heap, cover' → $*króu\dot{u}h_2\text{-}o\text{-}$ 'covering' > OCS *krovъ* 'roof').

5.3 The origin of $*kreu\dot{u}b^h\text{-}$

If we apply the comparandum that is $*p_lh_2(-)d^h h_1\text{-}\dot{\imath}e/o\text{-}$ to our $*kruh_2b^h\text{-}\dot{\imath}e/o\text{-}$ 'render covered, heap (upon)', the result is equally clear. Once the *Weather Rule* deleted the laryngeal, the sequence consisted of a suffix $*\text{-}\dot{\imath}e/o\text{-}$ and a base $*krub^h\text{-}$ interpretable as a root in its own right, namely $*kreu\dot{u}b^h\text{-}$ 'heap, cover'. At this point, we can revisit the thematic Tocharian B *krauptär* 'gathers' in light of its unambiguously athematic cognate *kroptär* in Tocharian A (class I pres., 3 pl. *kropäntär*, Malzahn 2010: 614–616). Given this discrepancy, it appears probable that the athematicity of the Tocharian A form is a retention (Jasanoff p.c.). Given, moreover, the likelihood of this retention and the possibility that *kraup-* and *krop-* reflect an *o*-grade, it is probable that Proto-Tocharian inherited the reflex of a *molō*-present (Jasanoff 2003: 70–79) $*króu\dot{u}b^h\text{-}/kréu\dot{u}b^h\text{-}$ 'heap up together' derivationally parallel to the τόμος-type $*króu\dot{u}b^h\text{-}o\text{-}$ 'accumulation' apparently ancestral to TB *kraupe* 'group'. In other words, once a $*kreu\dot{u}b^h\text{-}$ 'heap, cover' was posited on the basis of a historically secondary formation synchronically interpretable as primary, namely a $*\text{-}\dot{\imath}e/o\text{-}$ -present, the formation of primary stems, namely a *molō*-present and τόμος-type nominal, became possible.¹⁰

¹⁰ A similar account may be assigned to the preform of Proto-Germanic $*wal\dot{d}\text{-}i/a\text{-}$: Proto-Slavic $*vold\text{-}e/o\text{-}$: Proto-Baltic $*véld\text{-}$ 'be powerful, rule', which Rothstein-Dowden (2022: 129–135) has analyzed as a *molō*-present formed with suffixal $*\text{-}d^h\text{-}$ ($*(h_2)uólh_x\text{-}d^h\text{-}/*(h_2)uélh_x\text{-}d^h\text{-}$ 'be powerful').

6 The origin of **χreuðanq*

6.1 Introduction

A similar kind of root-based account is crucial to explaining the origin of secondary **kreuð^h*- and Proto-Germanic **χreuðanq* ‘deck, equip, adorn’. In light of our account of the origin of the secondary **kreuð^h*- in the anticausative phrase **kréh₂-u-h₁ b^huh_x-* ‘become covered’, it is reasonable to suppose that **kreuð^h*-, provisionally definable as ‘cover’, originated ultimately in the causative variant of the same construction, namely **kréh₂-u-h₁ d^heh₁-* ‘provide with covering’, on the basis of which a compound adjective **kréh₂-u-d^hh₁-ó-* was formed. Though this adjective could certainly convey the meaning ‘providing with covering’ as a function of the agentivity assignable by the causative light verb, the necessary existence of a patient/theme in the structure of **d^heh₁-* would always render possible the interpretation of **kréh₂-u-d^hh₁-ó-* as an adjective of patientive and resultative sense equivalent to its sister **kréh₂-u-b^huh_x-ó-* ‘(having) become heaped, covered’. We may thus suppose that **kréh₂-u-d^hh₁-ó-* also meant ‘provided with cover, rendered covered’.

6.2 COVERED FOR X

This patientive and resultative sense ‘covered’ is a nucleus disposed, via the basic notion of positioning one entity over and upon another entity, to generate by metonymy a variety of senses, some of which correspond approximately to Property Concepts. For example, the sense of Vedic *gúhā* ‘secret’ is, as discussed above, the product of COVERED FOR HIDDEN. The structurally and semantically interesting feature of this metonymy is its introduction of the end-oriented notion that the action relates to a third entity, which in this case is a non-experiencer, beside the patient/theme and implied agent. In other words, when a verb that means ‘cover’ is used in the sense ‘hide’, there arises the implication that the end or purpose of the agent’s action is to prevent the patient/theme from becoming a stimulus to some additional person (third entity), whose existence is implied, and which may even surface as an participant, as in (agent) *x*’s hiding of *y* (non-stimulus) from *z* (non-experiencer). The metonymy COVERED FOR HIDDEN that generates the implication of this third entity is structurally comparable to two other metonymies, COVERED FOR EQUIPPED and COVERED FOR ADORNED, which both imply ends, respectively, something for which the patient/theme is equipped and someone for whose experience the patient/theme is adorned. If we assume that **kréh₂-u-d^hh₁-ó-* ‘provided with cover, covered’ and its reflex **kruh₂(-)-d^hh₁-ó-* could develop senses of generic end-orientation and

even end-orientation in relation to an experiencer considered capable of assigning positive value to the stimulus (= patient/theme), it would be conceivable that **kruh₂(-)d^hh₁-ó-* expressed, on the one hand, the senses ‘equipped, prepared, ready’, and, on the other, ‘adorned, embellished, beautiful’. In this connection, it is appropriate to observe that Latin *parātus* ‘ready’ (: *parāre* ‘prepare’) and *ōrnātus* ‘adorned’ (: *ōrnāre* ‘fit out’), the resultative adjectives corresponding to these two sense-domains, have been lexicalized as adjectives bestriding the Activity and Property Conceptual domains. Correspondingly, we may suppose that **kruh₂(-)d^hh₁-ó-* meant something like ‘equipped, adorned’.

6.3 **kruh₂-d^hh₁-ó- → *kréuh₂(-)d^hh₁-o/es-*

The interaction of this quasi-Property Conceptual meaning and the behavior of the proto-language with respect to word-formation is now of great importance. Like its sister, the merely resultative **kruh₂b^h-ó-* ‘heaped, covered’, **kruh₂(-)d^hh₁-ó-* may be assumed to have been at some point analyzed as a simple thematic adjective **kruh₂d^hh₁-ó-*. Given, however, its developed and thus distinctive PC meaning, it is conceivable that speakers felt a greater need for that meaning to be expressed by means of an associated substantive. For the substantival expression of both Activity Concepts and Property Concepts, the readily available nominal stem formant was of course **-o/es-*, the addition of which was prototypically accompanied by the installation of radical *é*-grade for both primary and secondary derivation (e.g., **men-* ‘think, feel’ → **mén-o/es-* ‘mind, feeling’ > Ved. *mānas-* ‘spirit’; **sru-tó-* ‘flowing’ → **sréu-t-o/es-* [i.e., **sréu-to/e-s-*] ‘flow’ > Ved. *srótas-* n. ‘stream’). It is therefore reasonable to suppose that our **kruh₂d^hh₁-ó-* ‘decked, equipped, adorned’ was at some point the base of an *s*-stem substantive **kréuh₂(-)d^hh₁-o/es-* that expressed the resultative state of being equipped or adorned as a *nomen qualitatis* (cf. PDE *prepared-ness* ‘readiness’).

6.4 **χrusti- < *krud^h-s-ti-*

Before considering the influential role of this *s*-stem in our account, we should of course adduce evidence that such a stem existed in the first place. That evidence, while indirect, is all the same fairly clear. Old High German *hrust* ‘armor’ and OE *hyrst* ‘decoration, armor’, being exact cognates, point to a **χrusti-* derived from **χreuð-* by means of a suffix which Seebold (1970: 275) represents as **-sti-* (**hrud-sti- → *hrust-i-z* [f.] ‘Rüstung’). At face value, such a sequence is necessary to account for the *-st-* cluster actually attested, as we would expect an underlying Pre-Proto-Germanic **krud^h(h₁)-ti-* to be reflected as **χrussi-* (cf. OE *gewiss* ‘sure,

certain’ : **uid-tó*- ‘seen, known’). Moreover, though it is imaginable that **χrussi*- was refashioned as **χrusti*- or that the reflex of an original **krud^h-ti*- developed an excrescent -s- resulting in **χrusti*- as well, such claims would be ad hoc. It is therefore preferable to suppose that the preform was a **krud^h-s-ti*- of some description.

6.5 **χrausta*- < **kroud^h-s-to*- : **krud^h-s-tó*-

Since *-*ti*- in this non-primary context points to derivation from an adnominal in *-*tó*- (Schindler 1980: 390), a safe assumption would be that **krud^h-s-ti*- was derived from a **krud^h-s-tó*- (cf. Pre-Hitt. **dalugaš*- ‘length’ → **dalugašta*- ‘lengthy, long’ → **dalugašti*- ‘lengthiness, length’, Jasanoff 2004: 148). Indirect evidence for this **krud^h-s-tó*- may be derived from Old Norse *hraustr* ‘strong, valiant, doughty’, which certainly reflects a **χrausta*- related to **χreuðanq* (Hill 2003: 174), the lexical semantic basis being the connection between strength or vitality and preparedness, as evident in the relation between *rüstig* ‘healthy’ and *rüsten* ‘equip’, the reflex of a denominal to **χrusti*- (cf. OE *hyrstan* ‘adorn’). Formally, **χrausta*- ‘ready’ would thus reflect a **kroud^h-s-to*- interpretable as an adjective formed by insertion of *o*-grade into the root of **krud^h-s-tó*- (cf. **b^hoid-ro*- ‘cutting’ [Goth. *baitrs* ‘bitter’] : **b^hid-ró*- ‘id.’ [OE *bitter*, etc.]).

6.6 **kréud^hh₁-o/es*- ‘equipment, adornment’

If by comparison we consider, for example, the relation between (1) the thematic adjective basic to the preform of OE *āst* f. ‘kiln’ (< PGmc. **aistō*-) and (2) the *s*-stem that is ancestral to Vedic *édhas*- ‘fuel’ and basic to that thematic adjective (**h₂eǵd^h-s-to*- ‘hot’ ← **h₂eǵd^h-es*- ‘heat’, Nussbaum 1998: 527), it is appropriate to assert that the absolute zero-grade **krud^h-s-tó*- ‘equipped, adorned’ (< **krud^hh₁-s-tó*-)¹¹ was derived from an *s*-stem **kréud^hh₁-o/es*- ‘state of being equipped/adorned; equipment, adornment’, whose preform **kréuh₂d^hh₁-o/es*- reconstructed above had presumably undergone Hackstein’s (2002) rule of laryngeal deletion (*CH.CC > *C.CC). The question that now confronts us is the relation between this **kréud^hh₁-o/es*- and the preform of Proto-Germanic **χreuðanq*.

6.7 πλήθω (re-)revisited

This question is best addressed by considering the history of πλήθω ‘be/become full’, which LIV² 482–483 classifies as the reflex of a present in *-*d^he/o*-, namely

¹¹For loss of **h₁* between obstruents, see Jasanoff 2003: 77.

**pléh₁-d^he/o-*, also reflected in the diathetically divergent *frāda-* ‘increase (tr. act./intr. mid.)’ of Young and Old Avestan. What needs to be fundamentally explained in an account of the origin of this suffixal material is not this divergence in relation to some original semantic value of that material, but how the preform of *πληθ-* and *frād-* came to behave derivationally as a root when we know it is not a root at the deepest reconstructible stratum. The primary root was of course **pleh₁-* ‘fill’, which, like **kel-* ‘cover’ or **kreh₂-* ‘cover, heap’, corresponded to an Activity Concept, rather than a Property Concept (cf. Nussbaum 2022: 208–209). As such, **pleh₁-* was eligible to form a root noun **pléh₁-/plh₁-*’ with the value of a *nomen actionis* ‘filling’. Since, moreover, the specific lexical meaning of **pleh₁-* was of such a kind that the predication of **plh₁-éh₁* ‘with filling’ could express a state of being filled and thus full (cf. ‘with covering’ ≈ ‘covered’), the stative expression could be used in the causative periphrasis **plh₁-éh₁ d^heh₁-* ‘provide with filling, get *x* full’, on the basis of which was formed a compound **plh₁-d^hh₁-ó-* ‘filling; filled, full’. Like the resultative adjective **plh₁-nó-* ‘filled, full’ (Ved. *pūrṇá-* ‘full’) beside **pélh₁-n-o/es-* ‘fullness’ (: *parīṇas-* ‘fullness’, Av. *parānah-* ‘copious’, Rau 2009: 132, 174), **plh₁-d^hh₁-ó-* ‘filled, full’, indirect evidence for which may be derived from the reflexes of nominals belonging to clearly de-thematic categories (e.g., *ā*-stem Locrian *πληθα* ‘assembly, the commons’ < **pléh₁-d^hh₁-e-h₂-* ‘the full, fullness’, *u*-stem Lat. *plēbēs* ‘the commons’ < **pléh₁-d^hh₁-u-*; v. Nussbaum 2014), appears to have been the base of an *s*-stem **pléh₁-d^hh₁-o/es-* ‘fullness’ ancestral to *πληθος* (cf. **frādah-* in Av. personal name *Daḥhu-frādah-* ‘whose increase is of the land’, Yt.13.116, Bartholomae 1904: 682).

6.8 **pléh₁-d^hh₁-o/es-* ‘fullness’ : **pléh₁-d^hh₁-e/o-* ‘get full’

Since that *s*-stem was the substantival exponent of a Property Concept FULL, it could serve as a means of derivational inspiration on the model of (primary) adjectival roots interpretable as basic to *s*-stems and to prototypically radical *é*-grade simple thematic presents (e.g., **tep-* ‘hot’ → **tép-o/es-* ‘heat’ [> Ved. *tāpas-* ‘id.]: **tép-e/o-* ‘be/become hot’ [Ved. *tāpa-ti/te* ‘is/becomes hot’], **g^{uh}er-* ‘warm’ → **g^{uh}ér-o/es-* ‘warmth’ [θέρπος ‘summer’]: **g^{uh}ér-e/o-* ‘(be) warm’ [θέρομαι ‘become warm’], etc.). While the ultimate origin of the latter category is mysterious,¹² Rau (2013: 263–265; this volume) has observed what appears to be an archaic correlation of active morphology and intransitive semantics, which may point to the frequent – though not necessarily exclusive – anticausative use of the proto-middle. There may have been a degree of lability comparable to the

¹²On the thematization of the proto-middle, see Jasanoff 2023.

(anti)causative use of PDE *get* (e.g., *get* (x) *warm*; cf. PDE *become* : NHG *bekommen* ‘get’). In any case, given the cooccurrence of such Calandish derivatives, namely radical *é*-grade *s*-stem substantives and largely anticausative radical *é*-grade simple thematic presents, it is highly likely that **pléh₁-d^hh₁-o/es-* ‘fullness’ invited the suffixally substitutive formation of a **pléh₁-d^hh₁-e/o-* ‘get full’, whose Greek reflex is active and intransitive and whose Iranian one either happened to be active and transitive or was an original anticausative active remodeled as a middle corresponding to a causative active backformed from that middle in a manner suggested by Rau (2009: 155). However it might have been, the fundamental idea is that, for the purpose of derivation, the sequence **pleh₁d^hh₁-* was treated as a root with the meaning ‘filled, full’.

6.9 **-d^hh₁e/o-* and **-éh₁ie/o-* compared

Such an account is not designed to deny the existence of **-d^he/o-* presents with intransitive semantics. Indeed, Greek evidence makes it very clear that such a formant existed (φλεγέθω ‘burn’, φθινύθω ‘perish’, βαρύθω ‘am heavy’, φαέθω ‘shine’, μινύθω ‘diminish’; see Magni 2008: 268; Rothstein-Dowden 2022: 18–24). What may be suggested concerning the origin of the category of **-d^he/o-* presents is an inference based on cases like **pléh₁-d^hh₁-e/o-* ‘get full’, where the “root-based” formation from a base ending in **-d^hh₁-* was still analyzable with a morphological boundary before **-d^hh₁-*. Thus, a reanalyzed **-d^hh₁e/o-* could even be used to form primary verbal stems with intransitive meaning (e.g., **ieh₂-* : OCS *jadq* ‘go’ : Lith. *jóju*, *jóti* ‘id.’, **pelh₂-* : πελάθω/πλάθω ‘approach’ : πελάζω ‘id.’). In this respect, a largely anticausative **-d^hh₁e/o-* would, in terms of the history of the possessive adnominal system, intimately resemble the stative **-éh₁ie/o-*, which originated in the addition of **-ie/o-* to the instrumental of root noun abstracts used in nominal sentences (Jasanoff 2004: 147). Since these instrumental root nouns were usually predicated of themes or non-agents generally (e.g., **pédom tṛséh₁* ‘the ground is dry’, **h₃régōn h₂nréh₁* ‘the king is strong’, etc.),¹³ the historically complex suffix has a stative or anticausative value. Though in principle a **h₁rud^h-éh₁-ie/o-* might have been a causative (i.e., factitive) ‘reddden (x)’ like other adnominally based **-ie/o-* presents, such as what are arguably the preforms of κρύπτω and πλάσσω, the fact that it meant ‘be/become red’ is plausibly attributed to the use of **h₁rud^h-éh₁* ‘with redness’ in stative predication of themes.¹⁴ Like hypostatic **-ó-* and **-iio-* in nominals, **-ie/o-* could, as one of its functions,

¹³I use “theme” to refer an argument of which a property is predicated.

¹⁴For a formal analysis of the development of **-eh₁-* into a verbal stem formant, see Grestenberger 2023: 15–23.

merely render an already inflected substantive – and thus uninflectable adnominal – inflectable. Once a unitary **-éh₁ie/o-* ‘be/become (x)’ emerged, it could be used in primary derivation (e.g., Lat. *iaceō* ‘lie’ : *iaciō* ‘throw’). A comparable development would be the case for **-d^hh₁e/o-*, whose denominal origin from compounds based on the causative **d^heh₁-* periphrasis would follow from the greater importance and thus frequency of mentioning agents in causation, and whose largely intransitive semantics would be a consequence of the tendency for verbal adjectives to group patency and resultativity together for the sake of characterization (Haspelmath 1994). Because verbal adjectives in **-d^hh₁-ó-* were largely stative of patient/theme-oriented and resultative sense (**pl_hh₁-d^hh₁-ó-* ‘filled’), the adjectival abstracts formed from them on the basis of their resultative stative meaning (or secondary simple stative sense) could inspire the creation of largely intransitive, radical *é*-grade “simple” thematic presents formed from roots (or derivationally root-like bases) basic to Caland systems. Accordingly, the largely intransitive value of the suffix **-d^hh₁e/o-* would be an inheritance from the formation of which its immediate ancestor **R(é)(-d^hh₁-e/o-* was a subtype: radical *é*-grade simple thematic presents (**R(é)-e/o-*).

6.10 **kréud^hh₁-e/o-* ‘render equipped, adorned’

For our purpose, the question is how the **pléh₁(-d^hh₁-e/o-* ancestral to **pléh₁-d^hh₁e/o-* was formed. By appeal to an s-stem **pléh₁(-d^hh₁-o/es-* ‘fullness’ interpretable as derivative of a root-like base **pleh₁(-d^hh₁-* of Property Conceptual sense, the s-stem could easily have inspired a verbal formation in **-e/o-* (e.g., **g^{uh}er-* ‘warm’ → **g^{uh}ér-o/es-* ‘warmth’ : **g^{uh}ér-e/o-* ‘(be) warm’ :: **pléh₁(-d^hh₁-o/es-* ‘fullness’ : *x = *pléh₁(-d^hh₁-e/o-*; thus **pleh₁(-d^hh₁-* ‘full’ → **pléh₁(-d^hh₁-e/o-* ‘get full’). In the same way, I argue, our **kréud^hh₁-o/es-* ‘state of being equipped, adorned, equipment, adornment’, interpretable as derivative of a quasi-Property Conceptual **kreud^hh₁-* ‘equipped, adorned’, inspired the formation of a **kréud^hh₁-e/o-* ‘render equipped, adorned’ whose transitivity would be the result of whatever underlies the transitivity of the Avestan reflex of **pléh₁(-d^hh₁-e/o-* ‘get full’ (> *frāda-* ‘increase [tr. act./intr. mid.]’) and whose Proto-Germanic reflex **χreuðanq* (> OE *hréodan** ‘deck, adorn’) consequently came to mean ‘equip, adorn’.

7 Conclusion

The main purpose of this contribution has been the argument that, most notably among a group of formally and semantically similar verbal derivatives, Greek

κρύπτω ‘cover, hide, bury’ and Old English *hréodan** ‘deck, adorn’ are not only related in terms of root-etymology, but also paradigmatic of the two ways in which a verb can be de-adjectival: stem-derived (κρύπτω) vs. root-derived (*hréodan**). While I have in general focused on the morphological and derivational semantic foundations underlying the formation of the etyma of both verbs and their relatives, it has been my particular intention to describe the conceptual and lexical semantic basis for a type of root-based derivation that in IE terms may be designated as “de-adjectival”, namely Caland system verbal derivation (see the introduction to this volume). The formal prerequisite to this description has been the identification of the origin of the roots ancestral to those of κρύπτω and *hréodan** in nominal derivation and the predicative instrumental (anti)causative periphrasis – a type of construction closely, but not exclusively, associated with the Caland system and the actional expression of Property Concepts.

Despite the subsequent importance of Property Conceptuality in this contribution, I have argued that the primary root underlying all the forms compared was in lexical semantic terms a verbal one **kreh₂-* ‘heap, strew, cover’ (Latvian *krāju*, *krāt* ‘put aside, accumulate, amass’) corresponding to an Activity Concept and basic to a *u*-stem verbal noun **kró/éh₂-u-* ‘heap, covering’ whose instrumental was employed with anticausative **b^huh_x-* ‘become’ and causative **d^heh₁-* ‘put, make’. While the expression of an action involving this substantive as a possessum was in the present system handled by means of a derivative in **-ĵe/o-* (**kreh₂-u-ĵe/o-* ‘provide with covering’), whose reanalysis as a primary verb gave rise to a historically secondary verbal **kreu_hh₂-* (OCS *kryti* ‘cover’), constructions with the light verbs **d^heh₁-* and **b^huh_x-* were required elsewhere. I have argued that **k_ṛh₂-u-b^h(u)-ó-* ‘heaped, covered’ (> **kruh₂(-)^hó-*), a thematic compound formed on the basis of the latter construction, gave rise to the secondary verbal root **kreu_b^h-* (Tocharian B *kraup-* ‘gather, amass’) and that the origin of this root lies in stem-based de-adjectival derivation. In a manner comparable to the use of **-ĵe/o-* to represent the functionally adnominal **kréh₂-u-h₁* ‘with covering’ in the derivative **kreh₂-u-ĵe/o-* ‘provide with covering’, the inflectable adnominal **kruh₂(-)^hó-* ‘covered’ was the base of a **kruh₂^h-ĵe/o-* ‘render covered, heap (upon)’ whose reflex **krub^h-ĵe/o-* was the preform κρύπτω.

The crucial contrast in terms of de-adjectival derivation is with the preform of *hréodan** ‘deck, adorn’, which arguably originated in a similar way, namely, from a compound **k_ṛh₂-u-d^hh₁-ó-* ‘covered’ (> **kruh₂(-)^hh₁-ó-*) formed on the basis of the causative periphrasis **kréh₂-u-h₁ d^heh₁-* ‘provide with covering’, but whose ancestral root’s meaning emerged as a result of a lexical semantic development that endowed the resultative verbal adjective **kruh₂(-)^hh₁-ó-* ‘covered’ with the capacity to express an idea in the vicinity of a Property Concept, i.e.,

‘equipped, ready’ or ‘adorned’. This adjacency to Property Conceptuality was the precondition for the emergence of Calandish derivational behavior. Once **kruh₂(-)d^hh₁-ó-* ‘equipped, adorned’ became the base of an s-stem substantive **kréuh₂(-)d^hh₁-o/es-* ‘state of being equipped/adorned’, the resulting **kréud^hh₁-o/es-* ‘equipment, adornment’, now analyzed as a primary derivative, implied the existence of a derivative belonging to a verbal stem type closely associated with the Caland system: the radical é-grade simple thematic present (**kréud^hh₁-e/o-* ‘render equipped/adorned; provide with equipment/adornment’). In sum, the pre-forms of both Greek κρύπτω ‘cover’ and Proto-Germanic **χreuðanq* ‘deck, adorn’ not only exemplify the complexity of de-adjectival derivation, but also offer some insight into the ways in which morphology and derivational semantics interact with our conceptual world.

Abbreviations

act.	active	OE	Old English
Alb.	Albanian	OCS	Old Church Slavonic
Av.	Avestan	OE	Old English
f.	feminine	OHG	Old High German
Hitt.	Hittite	OIr.	Old Irish
Hom.	Homeric	OLith.	Old Lithuanian
Hsch.	Hesychius	ON	Old Norse
IE	Indo-European	PC	Property Concept
intr.	intransitive	PDE	Present-day English
Lat.	Latin	PGmc.	Proto-Germanic
Latv.	Latvian	PGk.	Proto-Greek
Lith.	Lithuanian	PIE	Proto-Indo-European
m.	masculine	TA	Tocharian A
mid.	middle	TB	Tocharian B
n.	neuter	tr.	transitive
NHG	New High German	Ved.	Vedic
OCS	Old Church Slavonic	Yt.	Yašt

References

- Adams, Douglas Q. 2013. *A dictionary of Tocharian B*. 2nd ed. Amsterdam: Rodopi.
 Balles, Irene. 2006. *Die altindische Cvi-Konstruktion. Form, Funktion, Ursprung*.
 Bremen: Hempen.

- Bartholomae, Christian. 1904. *Altiranisches Wörterbuch*. Strassburg: Trübner.
- Buck, Carl Darling. 1949. *A dictionary of selected synonyms in the principal Indo-European languages*. Chicago: University of Chicago Press.
- Catsanicos, Jean. 1985. Review of Jos Weitenberg, *Die hethitischen u-Stämme*. *Bulletin de la Société Linguistique de Paris* 80(2). 123–127.
- Dixon, R. M. W. 1982. *Where have all the adjectives gone? And other essays in semantics and syntax*. The Hague: Mouton.
- Dixon, R. M. W. 2004. Adjective classes in typological perspective. In R. M. W. Dixon & Alexandra Y. Aikhenvald (eds.), *Adjective classes: A cross-linguistic typology*, 1–49. Oxford: Oxford University Press.
- EDG = Beekes, Robert. 2010. *Etymological dictionary of Greek*. Leiden: Brill.
- EDL = de Vaan, Michiel. 2008. *Etymological dictionary of Latin and the other Italic languages*. Leiden: Brill.
- Francez, Itamar & Andrew Koontz-Garboden. 2015. Semantic variation and the grammar of property concepts. *Language* 91(3). 533–563. DOI: 10.1353/lan.2015.0047.
- Grestenberger, Laura. 2023. The diachrony of verbalizers in Indo-European: Where does *v* come from? *Journal of Historical Syntax* 7. 1–40. DOI: 10.18148/hs/2023.v7i6-19.156.
- Hackstein, Olav. 2002. Uridg. *CH.CC > *C.CC. *Historische Sprachforschung* 115. 1–22.
- Hackstein, Olav. 2021. On Latin *arbor* and why *tree* is grammatically feminine in PIE. In Hannes A. Fellner, Melanie Malzahn & Michaël Peyrot (eds.), *Lyuke wmer ra. Indo-European Studies in Honor of Georges-Jean Pinault*, 183–204. Ann Arbor: Beech Stave.
- Haspelmath, Martin. 1994. Passive participles across languages. In Barbara A. Fox & Paul J. Hopper (eds.), *Voice: Form and function*, 151–177. Amsterdam: Benjamins.
- Hill, Eugen. 2003. *Untersuchungen zum inneren Sandhi des Indogermanischen: Der Zusammenstoss von Dentalplosiven im Indoiranischen, Germanischen, Italischen und Keltischen*. Bremen: Hempen.
- Höfler, Stefan. 2017. Observations on the *palma* rule. *Pallas* 103. 15–23.
- IEW = Pokorny, Julius. 1959–1969. *Indogermanisches etymologisches Wörterbuch*. Bern: Francke.
- Jasanoff, Jay H. 1978. *Stative and middle in Indo-European*. Innsbruck: Institut für Sprachwissenschaft der Universität Innsbruck.
- Jasanoff, Jay H. 2003. *Hittite and the Indo-European verb*. Oxford: Oxford University Press.

- Jasanoff, Jay H. 2004. “Stative” *-ē- revisited. *Die Sprache* 43 (2002-03 [2004]). 127–170.
- Jasanoff, Jay H. 2023. PIE *g^wih₃ue/o- ‘live’, u-presents, and the prehistory of the thematic conjugation. *Die Sprache* 55 (2022-23 [2023]). 61–81.
- Karulis, Konstantīns. 1992. *Latviešu etimoloģijas vārdnīca*. 2nd ed. Rīga: Avots.
- Kloekhorst, Alwin. 2006. Initial laryngeals in Anatolian. *Historische Sprachforschung* 119. 77–108.
- Kloekhorst, Alwin. 2008. *Etymological dictionary of the Hittite inherited lexicon*. Leiden: Brill.
- LIV² = Rix, Helmut. 2001. *Lexikon der indogermanischen Verben*. Bearbeitet von Martin Kümmel, Thomas Zehnder, Reiner Lipp und Brigitte Schirmer. 2nd ed. Wiesbaden: Reichert.
- Magni, Elisabetta. 2008. Continuità e contiguità nelle categorie verbali: Le forme in -θ- del greco. *Archivio Glottologico Italiano* 93. 171–255.
- Malzahn, Melanie. 2010. *The Tocharian verbal system*. Leiden: Brill.
- Meiser, Gerhard. 1998. *Historische Laut- und Formenlehre der lateinischen Sprache*. Darmstadt: Heidelberg.
- Melchert, H. Craig. 1998. Hittite arku- “chant, intone” vs. arkuwā(i)- “make a plea”. *Journal of Cuneiform Studies* 50. 45–51.
- Melchert, H. Craig. 2013. Hittite hi-verbs of the type -āC₁i, -aC₁C₁anzi. *Indogermanische Forschungen* 117. 173–186.
- Merritt, Andrew. 2021. κέλῦφος and καλύπτω. *Indogermanische Forschungen* 126. 305–324.
- Neri, Sergio. 2017. *Wetter: Etymologie und Lautgesetz*. Università degli Studi di Perugia. http://www.ctl.unipg.it/issues/CTL_14.pdf.
- Nussbaum, Alan J. 1996. *Latin acētum, acūtus, aurītus, avītus: Four of a kind?* Paper presented at the 15th East Coast Indo-European Conference, Yale University, 14 June 1996.
- Nussbaum, Alan J. 1998. Severe problems. In Jay H. Jasanoff, H. Craig Melchert & Lisi Oliver (eds.), *Mír curad: Studies in honor of Calvert Watkins*, 521–538. Innsbruck: Institut für Sprachwissenschaft der Universität Innsbruck.
- Nussbaum, Alan J. 2004. A -t- party: Various IE nominal stems in *(o/e)t-. Paper delivered at the sixteenth Annual UCLA Indo-European Conference, November 5–6, 2004.
- Nussbaum, Alan J. 2014. The PIE proprietor and his goods. In H. Craig Melchert, Elisabeth Rieken & Thomas Steer (eds.), *Munus Amicitiae: Norbert Oettinger a collegis et amicis dicatum*, 228–254. Ann Arbor: Beech Stave.

- Nussbaum, Alan J. 2022. Derivational properties of “adjectival roots” (expanded handout). In Melanie Malzahn, Hannes Fellner & Theresa-Susanna Illes (eds.), *Zurück zur Wurzel—Struktur, Funktion und Semantik der Wurzel im Indogermanischen. Akten der 15. Fachtagung der Indogermanischen Gesellschaft vom 13. bis 16. September 2016 in Wien*, 205–224. Wiesbaden: Reichert.
- Rasmussen, Jens Elmegård. 1989. *Studien zur Morphophonemik der indogermanischen Grundsprache*. Innsbruck: Institut für Sprachwissenschaft der Universität Innsbruck.
- Rau, Jeremy. 2009. *Indo-European nominal morphology: The decads and the Caland system*. Innsbruck: Institut für Sprachen und Literaturen der Universität Innsbruck.
- Rau, Jeremy. 2013. Notes on state-oriented verbal roots, the Caland system, and primary verb morphology in Indo-Iranian and Indo-European. In Adam I. Cooper, Jeremy Rau & Michael Weiss (eds.), *Multi nominis grammaticus: Studies in Classical and Indo-European linguistics in honor of Alan J. Nussbaum on the occasion of his sixty-fifth birthday*, 255–273. Ann Arbor: Beech Stave.
- Rix, Helmut. 1996. Review of “Peter Schrijver, *The reflexes of the Proto-Indo-European larygeals in Latin*”. *Kratylos* 41. 153–163.
- Rothstein-Dowden, Zachary. 2022. *Dental-aspirate presents in Greek and Indo-European*. Cambridge, MA: Harvard University. (Doctoral dissertation).
- Schindler, Jochem. 1980. Zur Herkunft der altindischen *cvi*-Bildungen. In Manfred Mayrhofer, Martin Peters & Oskar E. Pfeiffer (eds.), *Lautgeschichte und Etymologie. Akten der VI Fachtagung der indogermanischen Gesellschaft. Wien, 24.–29. September 1978*, 386–393. Wiesbaden: Reichert.
- Seebold, Elmar. 1970. *Vergleichendes und etymologisches Wörterbuch der germanischen starken Verben*. Berlin: Mouton.
- Smoczyński, Wojciech. 2018. *Lithuanian etymological dictionary*. Berlin: Peter Lang.
- Villanueva Svensson, Miguel. 2014. Tone variation in the Baltic *ia*-presents. *Indogermanische Forschungen* 119. 227–249.
- Weiss, Michael. 2020. *Outline of the historical and comparative grammar of Latin*. 2nd ed. Ann Arbor: Beech Stave.

